$S = [+QK^T \rightarrow]$

$A = \text{softmax}(S) = \frac{e^{-Si}}{\sum_{i=1}^{n} si} = \frac{f}{di} * e^{-Si}$

$A = \frac{exp(Si - max(S))}{\sum_{i=1}^{n} exp(si - max(S))}$

$\rightarrow Mn = \max_{1 \le i \le n} Si$

$di = \sum_{i=1}^{n} e^{Si - max(S)}$ $\rightarrow$ $di = \sum_{i=1}^{n} e^{si - Mn}$

$\leftarrow QK^T \rightarrow]$

$t = i-1$

$M_{i-1} = \max_{1 \le j \le i-1} Sj$

$d_{i-1} = \sum_{j=1}^{i-1} e^{Sj - M_{i-1}}$

$t = i$

$M_i = \max_{1 \le j \le i} Sj$

$di = \sum_{j=1}^{i} e^{Sj - Mi}$

$di = \sum_{j=1}^{i-1} e^{Sj - Mi} + e^{Si - Mi}$

$di = \sum_{j=1}^{i-1} \frac{e^{M_{i-1}}}{e^{M_{i-1}}} e^{Sj - Mi} + e^{Si - Mi}$

$di = \sum_{j=1}^{i-1} \frac{e^{M_{i-1}}}{e^{Mi}} e^{Sj - M_{i-1}} + e^{Si - Mi}$

$di = \left(\sum_{j=1}^{i-1} e^{Sj - M_{i-1}} \right) \frac{e^{M_{i-1}}}{e^{Mi}} + e^{Si - Mi}$

$di = d_{i-1} * e^{M_{i-1} - Mi} + e^{Si - Mi}$

$S = QK^T$

$d_i = d_{i-1} * e^{M_{i-1} - Mi} + e^{Si - Mi}$

$\rightarrow$ Next $O = A \cdot V$

$A = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}_{2x2}$, $B = \begin{bmatrix} 5 & 6 & 7 & 8 \\ 9 & 10 & 11 & 12 \end{bmatrix}_{2x3}$

$O_{2x3} = A_{2x2} \cdot B_{2x3}$

$O_1 = [1, 2] \begin{bmatrix} 5 & 6 & 7 & 8 \\ 9 & 10 & 11 & 12 \end{bmatrix}$ ← array

$O_1 = 1[5, 6, 7, 8] + 2[9, 10, 11, 12]$

$O_{ij} = \sum_{k=1}^{2} A_{i,k} \cdot B_{k,j}$

$$O_{i,j} = A_{i,1} B_{1,j} + A_{i,2} \cdot B_{2,j}$$

↳ partial sum

$$S_{i,j}^{k} = \sum_{\epsilon=1}^{k} A_{i,\epsilon} B_{\epsilon,j}$$

$$S_{i,j}^{k} = \sum_{\epsilon=1}^{k-1} A_{i,\epsilon} \cdot B_{\epsilon,j} + A_{i,k} \cdot B_{k,j}$$

→ same we for

$$O = A V$$

$$O_{i,j} = \sum_{\epsilon=1}^{k} A_{i,k} \cdot V_{k,j}$$

↓ put the value of A ↑ softmax value.

$$d_i = d_{i-1} e^{M_{i-1} - M_i} + e^{x_i - M_i}$$

$$a_i = \frac{e^{s_i - M_n}}{d_i} \cdot v$$

$$\rightarrow M_{i-1} = \max_{1 \le j \le i-1} (s_i)$$

$$\rightarrow d_{i-1} = \sum_{j=1}^{i-1} e^{s_j - M_{i-1}}$$

$$\rightarrow o_{i-1} = \sum_{j=1}^{i-1} \frac{e^{s_j - M_{i-1}}}{d_{i-1}} \cdot v_j$$

$$M_i = \max_{1 \le j \le i} (s_j)$$

$$d_i = d_{i-1} e^{M_{i-1} - M_i} + e^{s_i - M_i}$$

$$o_i = \sum_{j=1}^{i} \frac{e^{s_j - M_i}}{d_i} \cdot v_j$$

$$o_i = \sum_{j=1}^{i-1} \frac{e^{s_j - M_i}}{d_i} v_j + \frac{e^{s_i - M_i}}{d_i} v_i$$

$$o_i = \sum_{j=1}^{i-1} \frac{e^{s_j - M_i}}{d_i} \left(\frac{d_{i-1}}{d_{i-1}}\right) \left(\frac{e^{M_{i-1}}}{e^{M_{i-1}}}\right) v_j + \frac{e^{s_i - M_i}}{d_i} v_i$$

$$o_i = \sum_{j=1}^{i-1} \frac{e^{s_j - M_{i-1}}}{d_{i-1}} \cdot \left(\frac{e^{M_{i-1}}}{e^{M_i}}\right) d_{i-1} \cdot \frac{e^{M_{i-1} - M_i}}{d_i} + \frac{e^{s_i - M_i}}{d_i} v_i$$

$$o_i = o_{i-1} * \frac{d_{i-1}}{d_i} * e^{M_{i-1} - M_i} + \frac{e^{s_i - M_i}}{d_i} v_i$$